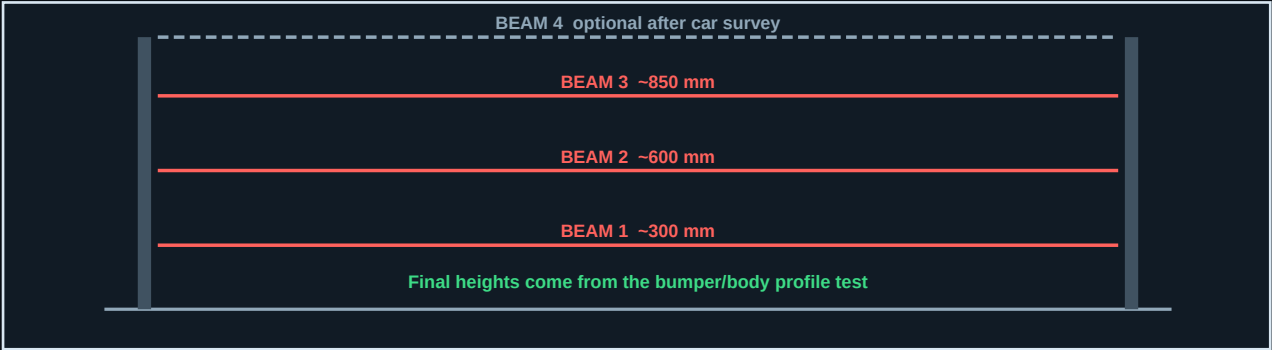


Mounting overview - fixed hardware, optical plane and driver indication

TOP VIEW - NOT TO SCALE



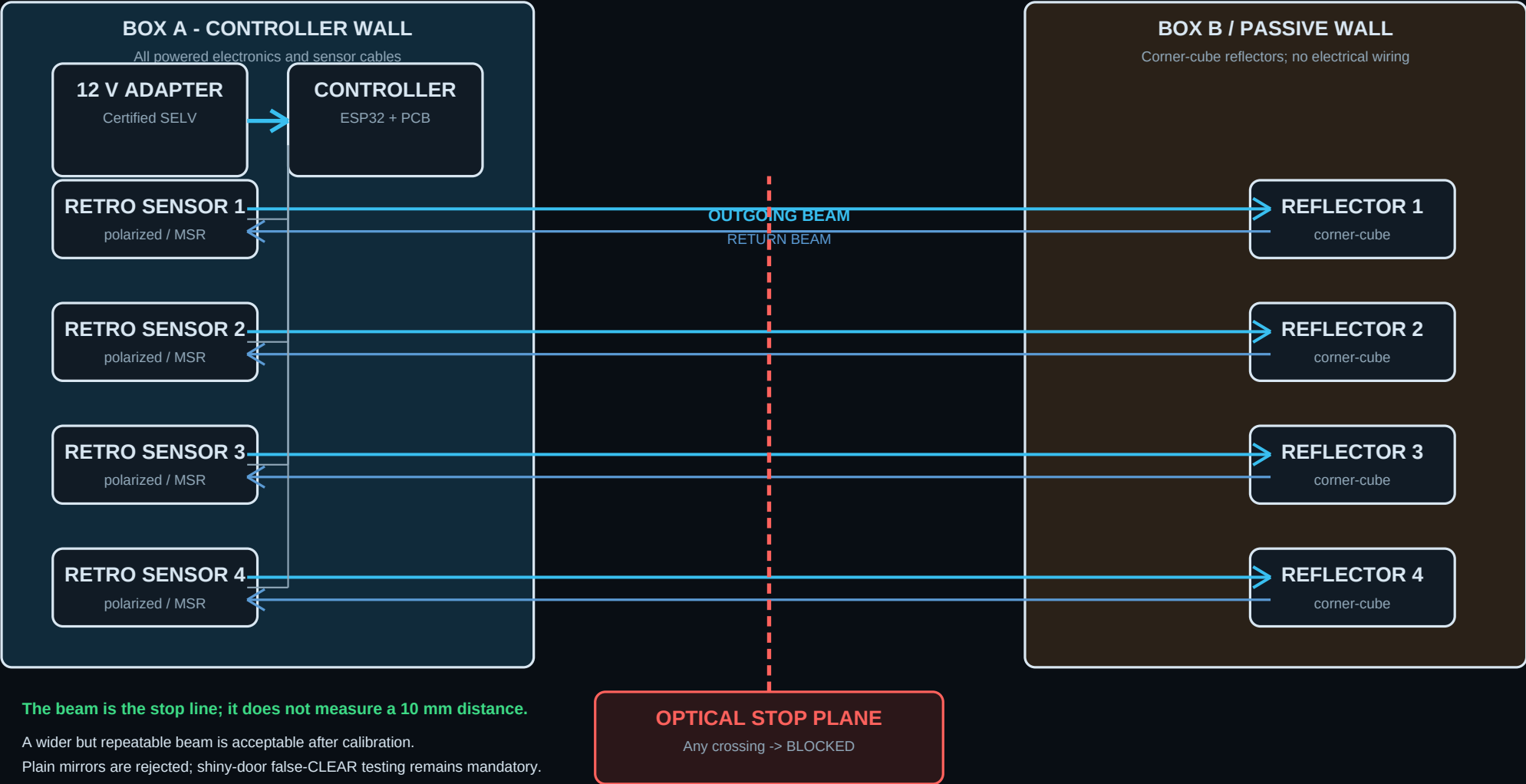
FRONT ELEVATION - EXPERIMENTAL BEAM HEIGHTS



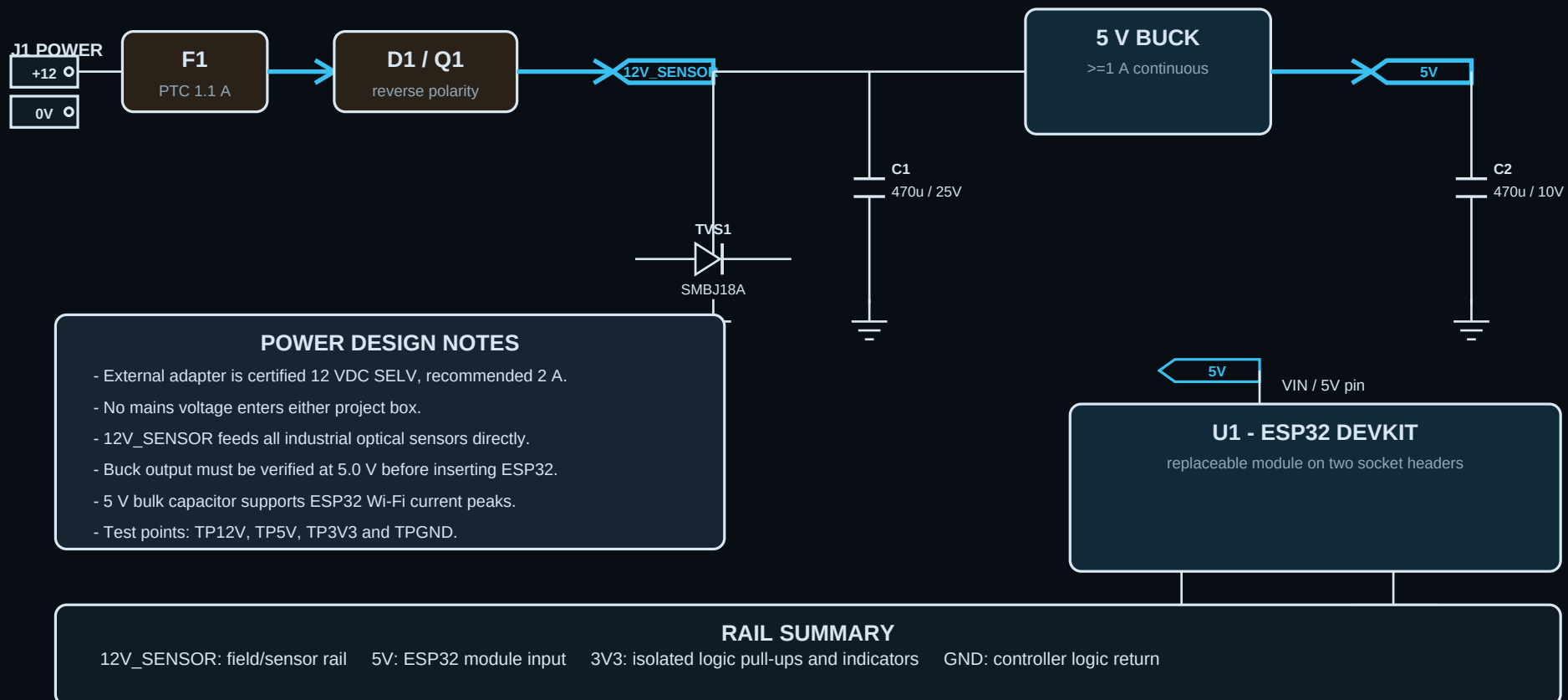
INSTALLATION RULES

1. Mount only to fixed masonry/reveal - never the door.
2. Keep optics clear of tracks, seals and moving hardware.
3. Use pan/tilt fine adjustment, then lock both axes.
4. Route 12 V SELV cable high; add drip loops/glands.
5. Use corner-cube reflector, not a plastic plain mirror.
6. Test shiny-door false CLEAR before trusting GREEN.
7. Measure door-to-car geometry before drilling holes.

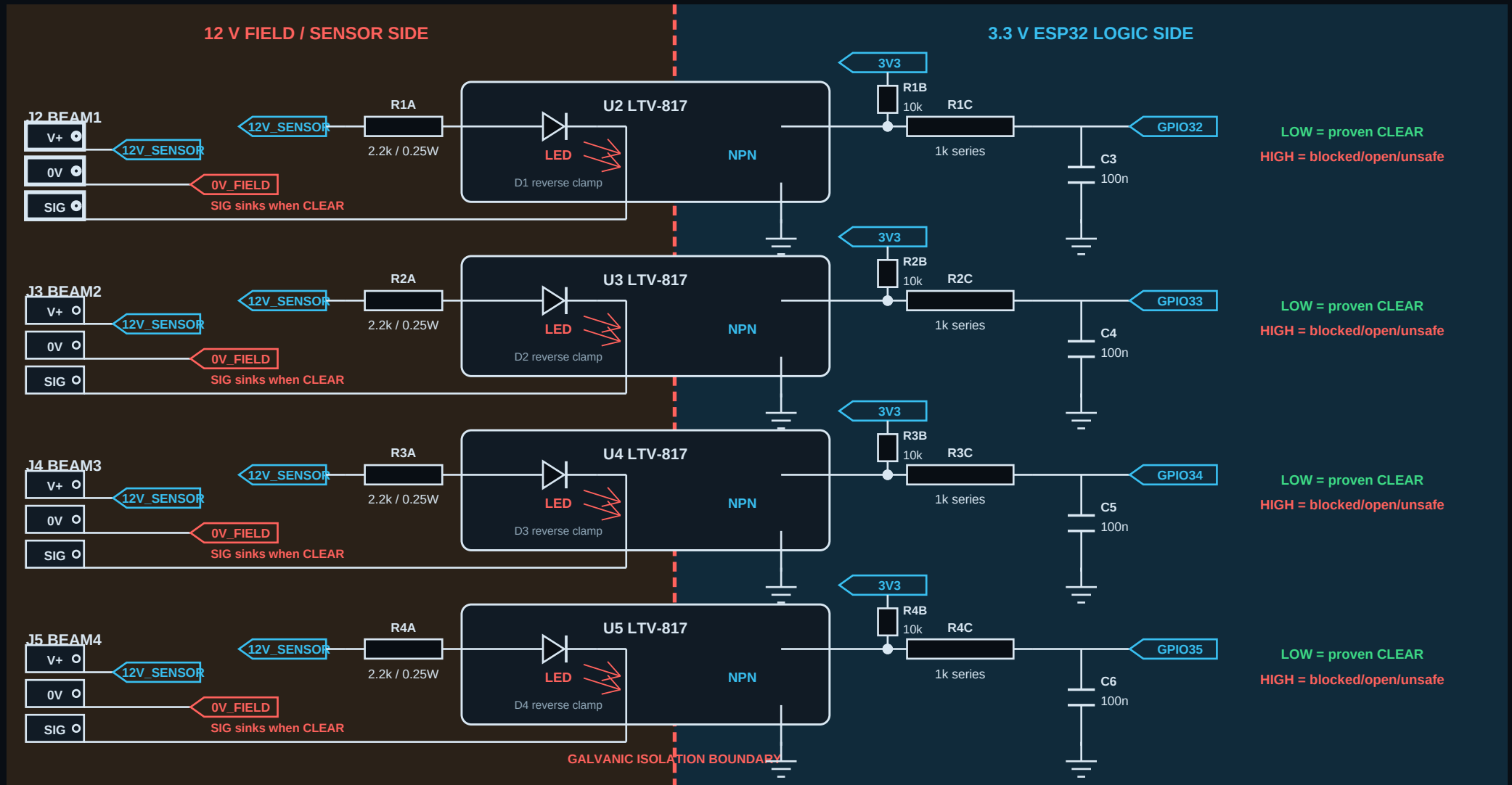
System architecture - fixed polarized retroreflective trip lines



Power input, protected rails and ESP32 module

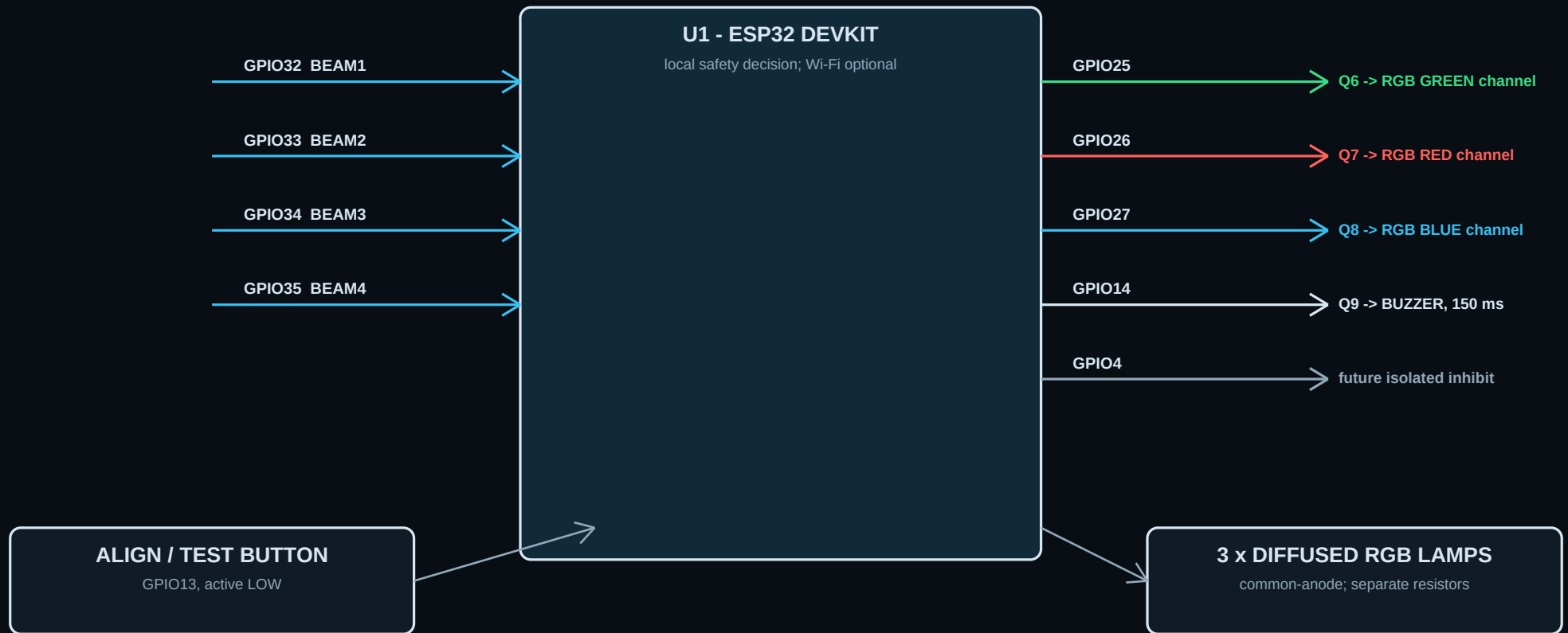


Four isolated fail-safe sensor inputs



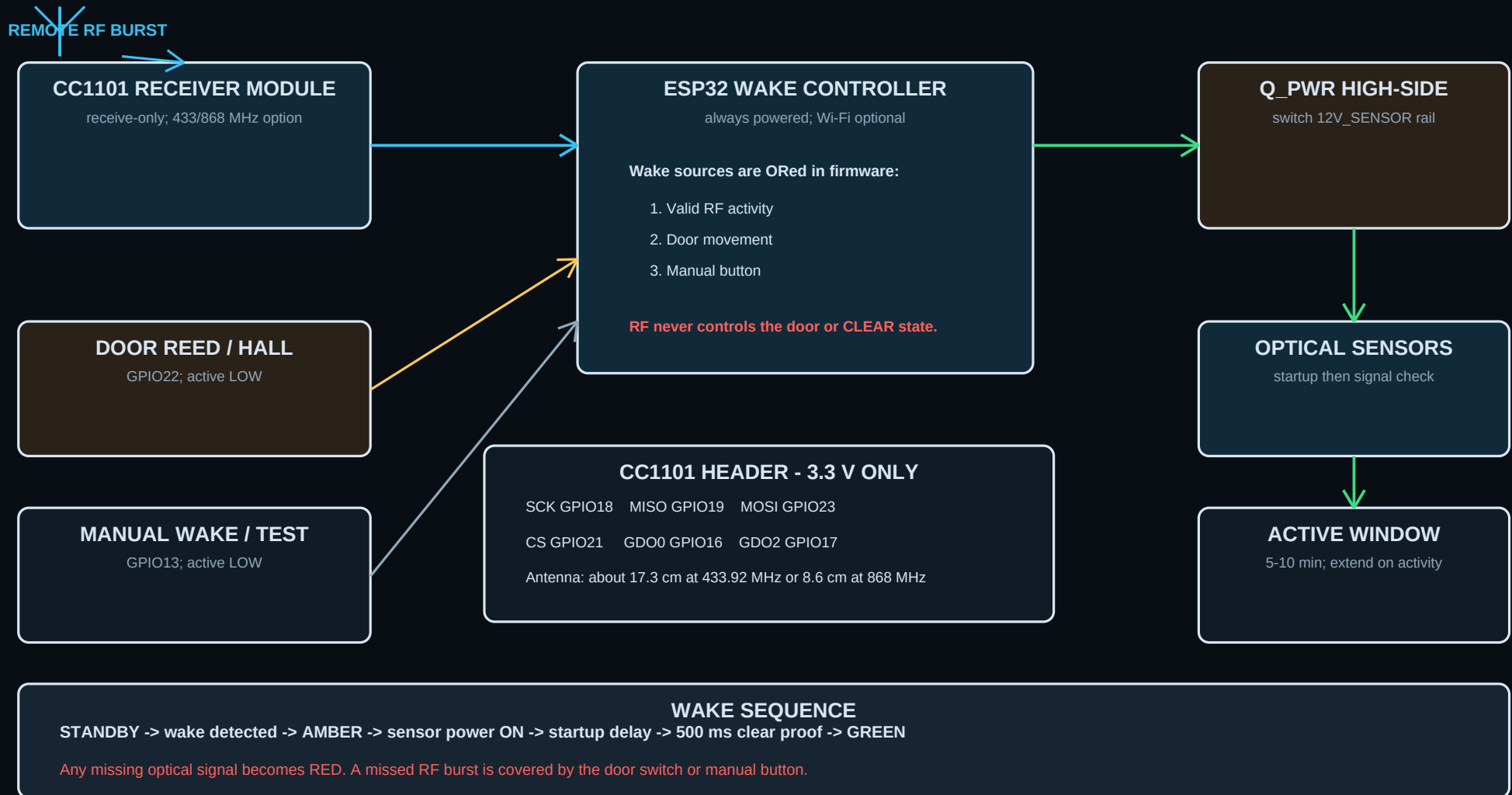
D2 behavior: blocked beam, open cable, dead transmitter and dead/open sensor are all the same RED / NOT CLEAR state.

ESP32 connections, redundant RGB indication and safety behavior



GREEN=clear. RED=blocked/no signal/dead sensor. AMBER=boot/unknown (red+green channels). Dark means no power.

Optional RF, door-motion and manual wake system



Prototype purchasing BOM - buy one beam first, then scale only after validation

Clickable BUY/SEARCH buttons open AliExpress searches. Prices are rough delivered-EU ranges and can change.

Critical rule: the cheap optical sensor is a test article. It earns GREEN only after the repeatability and shiny-door tests.

QTY	ITEM	MINIMUM DESCRIPTION	BUDGET	LINK	QTY	ITEM	MINIMUM DESCRIPTION	BUDGET	LINK
1	Laser retroreflective NPN sensor	12-24 V, polarized/MSR, reflector	EUR 15-35	ALIEXPRESS	1	Screw terminal kit	5.08 mm, 2-way and 3-way	EUR 4-10	ALIEXPRESS
1	Corner-cube reflector	Use supplied unit; spare 40-60 mm	EUR 3-10	ALIEXPRESS	3	Diffused common-anode RGB LED	5 mm plus 3 NPN driver transistors	EUR 2-6	ALIEXPRESS
1	ESP32 DevKit V1	ESP32-WROOM, 30-pin preferred	owned / 4-9	ALIEXPRESS	1	Active buzzer + push button	3-5 V buzzer; momentary test switch	EUR 2-5	ALIEXPRESS
1	LM2596 buck module	12 V to 5.0 V, set before ESP32	EUR 1-4	ALIEXPRESS	1	CC1101 receiver (optional)	Match remote: 433.92 or 868 MHz	EUR 3-9	ALIEXPRESS
4	LTV-817 / PC817 optocoupler	DIP-4; four channels	EUR 1-3	ALIEXPRESS	1	Door reed switch (recommended)	Normally closed, wired backup wake	EUR 2-6	ALIEXPRESS
1	Resistor assortment	2.2k, 10k, 1k + LED resistors	EUR 3-8	ALIEXPRESS	1	IP65 ABS controller enclosure	Approx. 160 x 120 x 75 mm	EUR 8-18	ALIEXPRESS
1	Diode assortment	1N4148 plus protection spares	EUR 1-4	ALIEXPRESS	1	Passive reflector enclosure	Approx. 120 x 80 x 55 mm	EUR 5-12	ALIEXPRESS
1	Capacitor assortment	100 nF, 470 uF/25 V, 470 uF/10 V	EUR 4-10	ALIEXPRESS	1	Cable glands + low-voltage cable	M12/M16 glands; 3-core sensor cable	EUR 8-20	ALIEXPRESS
1	12 V protection set	SMBJ18A TVS + MF-R110 polyfuse	EUR 2-6	ALIEXPRESS	2	Adjustable camera-style brackets	Metal mini pan/tilt, lockable	EUR 6-18	ALIEXPRESS

BUY LOCALLY / EU DISTRIBUTOR: certified 12 VDC 2 A SELV adapter (e.g. Mean Well GST25E12-P1J). Do not put mains on the PCB.

Prototype total excluding sensor and owned ESP32: roughly EUR 45-105. Buy only ONE sensor/reflector until the optical boundary passes.

TME / 12 V PSU